

# Hydraulic System Maintenance Report

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## Executive Summary

Total cycles analysed: 5  
Cycles requiring attention: 4  
Healthy cycles: 1  
Overall system status: ACTION REQUIRED

## System Health Overview

Pump condition: Severe leakage  
Accumulator condition: Optimal pressure

## Cycle 2 - Detailed Finding

### Pump Leakage Probabilities:

|                |                        |     |
|----------------|------------------------|-----|
| No leakage     | <div><div></div></div> | 2%  |
| Weak leakage   | <div><div></div></div> | 92% |
| Severe leakage | <div><div></div></div> | 6%  |

### Accumulator Condition Probabilities:

|                           |                        |     |
|---------------------------|------------------------|-----|
| Close to failure          | <div><div></div></div> | 0%  |
| Severely reduced pressure | <div><div></div></div> | 0%  |
| Slightly reduced pressure | <div><div></div></div> | 2%  |
| Optimal pressure          | <div><div></div></div> | 98% |

### Top Contributing Sensors:

- [+] System Efficiency (Average) - high impact (value: 57.6486)
- [+] System Efficiency (Variability) - high impact (value: 22.8378)
- [+] Volume Flow 1 (Average) - high impact (value: 6.3483)

### AI Maintenance Finding:

#### FINDING:

The hydraulic system has shown weak leakage confirmed at 92% severity, indicating a potential problem with one of the pumps in the system.

#### SENSOR EVIDENCE:

The system efficiency sensors are showing high average and high variability, which can indicate a leak or air in the system. The volume flow 1 sensor is also high, which could be caused by the weak leakage. Motor power is moderate, but the overall system efficiency is still a concern.

#### CONSEQUENCE OF DEFERRAL:

If left unchecked, the weak leakage can lead to increased pump wear and potentially cause a more significant system failure.

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RECOMMENDED ACTION:

Schedule an immediate leak-down test on the pump in question during the next scheduled maintenance window to identify and repair the leak as soon as possible.

Cycle 3 - Detailed Finding

Pump Leakage Probabilities:

|                |             |     |
|----------------|-------------|-----|
| No leakage     | <div></div> | 0%  |
| Weak leakage   | <div></div> | 98% |
| Severe leakage | <div></div> | 2%  |

Accumulator Condition Probabilities:

|                           |             |     |
|---------------------------|-------------|-----|
| Close to failure          | <div></div> | 0%  |
| Severely reduced pressure | <div></div> | 0%  |
| Slightly reduced pressure | <div></div> | 1%  |
| Optimal pressure          | <div></div> | 99% |

Top Contributing Sensors:

- [+] System Efficiency (Average) - high impact (value: 56.6581)
- [+] System Efficiency (Variability) - high impact (value: 23.3455)
- [+] Volume Flow 1 (Average) - high impact (value: 6.3091)

AI Maintenance Finding:

FINDING:

A weak leakage in the pump system has been detected, classified as 'Weak leakage' with a high confidence level.

SENSOR EVIDENCE:

The system efficiency sensor readings are of particular concern. The average system efficiency is only 56.6581%, indicating the system is operating far from its optimal efficiency. Additionally, the high variability in system efficiency, 23.3455%, suggests that pressures may not be stable, which may also be contributing to the poor performance of the pump.

CONSEQUENCE OF DEFERRAL:

If left unaddressed, the weak leakage and inefficiencies in the system could lead to a significant decrease in overall system performance, reduced safety, and potentially cause premature wear on equipment.

RECOMMENDED ACTION:

Immediate action is required to investigate and address the weak pump leakage, including visual inspection, and to review and optimize system efficiency settings to bring them as close to optimal as possible.

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## Cycle 4 - Detailed Finding

### Pump Leakage Probabilities:



### Accumulator Condition Probabilities:



### Top Contributing Sensors:

- [+] System Efficiency (Variability) - high impact (value: 22.5673)
- [+] Volume Flow 1 (Average) - high impact (value: 6.3054)
- [+] Volume Flow 1 (Variability) - high impact (value: 2.8525)

### AI Maintenance Finding:

#### FINDING:

Severe leakage in the hydraulic pump system has been detected.

#### SENSOR EVIDENCE:

System efficiency and volume flow sensors flagged the issue, reading unusually high and variable values. These readings suggest that some fluid is escaping, resulting in lower-than-expected pressures and reduced system performance. High volume flow values further indicate that more pressure is being lost and less energy is being transferred.

#### CONSEQUENCE OF DEFERRAL:

If this issue remains unresolved, it can lead to increased wear on pump components, reduced overall system efficiency, and potentially cause a system failure during critical operations.

#### RECOMMENDED ACTION:

Inspect the pump system immediately to locate and repair or replace any leaking components, ideally within the current shift to prevent potential system shutdown or further degradation.

## Cycle 5 - Detailed Finding

### Pump Leakage Probabilities:



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Severe leakage94%

Accumulator Condition Probabilities:

Close to failure2%

Severely reduced pressure1%

Slightly reduced pressure11%

Optimal pressure86%

Top Contributing Sensors:

- [+] Volume Flow 1 (Average) - high impact (value: 3.0191)
- [+] System Efficiency (Average) - high impact (value: 27.1264)
- [+] Motor Power (Average) - high impact (value: 2676.6586)

AI Maintenance Finding:

FINDING:  
A severe leakage has been detected in the hydraulic system pump, posing a risk to system performance and potentially leading to equipment damage.

SENSOR EVIDENCE:  
Sensor evidence points to increased flow rates through the pump, which can cause excessive wear and tear on system components. The system's motor power and efficiency are also impacted, indicating the pump is working harder to maintain its output. These readings suggest the leakage is significant enough to compromise the system's overall efficiency.

CONSEQUENCE OF DEFERRAL:  
If left unaddressed, the severe leakage can lead to premature equipment failure, reducing system reliability and potentially causing costly downtime.

RECOMMENDED ACTION:  
Immediately inspect the pump for the source of the leakage and repair or replace the faulty component to prevent further system degradation.

Regulatory Reference

- ISO 4413 - Hydraulic fluid power: General rules and safety requirements
- API 686 - Recommended Practice for Machinery Installation and Installation Design
- ISO 18435 - Diagnostics and prognostics for hydraulic systems

All maintenance actions should be documented in the facility CMMS and approved by the maintenance superintendent before execution.